Problem templates

1 First-order lin. difference eq. with c.c. $\boxed{x_{k+1} + a x_k = b_k}$

→ general sol: $x_{k+1} + a x_k = 0$

$r + a = 0$

$r = -a \longmapsto (-a)^k$

$x_k = c \cdot (-a)^k, \ c \in \mathbb{R}.$

→ particular sol: of $x_{k+1} + a x_k = b_k$.

ex: $x_{k+1} + 3x_k = 2$

$a + 3a = 2 \Rightarrow 4a = 2 \Rightarrow a = \frac{1}{2} = x_p$ (part. sol.)

→ general (final) sol: $x = x_h + x_p$.

2 Second-order lin. diff. eq. with c.c. $\boxed{x_{k+2} + a \cdot x_{k+1} + b \cdot x_k = 0}$

→ general sol: $x_{k+2} + a x_{k+1} + b x_k = 0$

$r^2 + ar + b = 0$

$\Rightarrow r_1, r_2$

- $r_1 \neq r_2 \in \mathbb{R} \Rightarrow \longmapsto (r_1)^k, (r_2)^k$
- $r_1, r_2 \in \mathbb{C} \Rightarrow Re(\alpha + i\beta)^k, \ Im(\alpha + i\beta)^k$; $\alpha, \beta \in \mathbb{R}, \beta \neq 0.$
- $r_1 = r_2 \in \mathbb{R} \Rightarrow \longmapsto (r_1)^k, \ k \cdot (r_1)^k.$

$z = \alpha + i\beta = \rho(\cos\theta + i \cdot \sin\theta)$

$\rho = \sqrt{\alpha^2 + \beta^2}$

$\cos\theta = \frac{\alpha}{\rho}, \quad \sin\theta = \frac{\beta}{\rho}$

→ general (final) sol: $x = c_1 \cdot x^1 + c_2 \cdot x^2, \quad c_1, c_2 \in \mathbb{R}$

sequences found before

$x^1; x^2 : \mathbb{N} \to \mathbb{R}$.

(1)

Ex: 1. a) find sol. of the form $x_k = a \cdot 3^k$ of the eq. $x_{k+1} + (-2) x_k = 3^k, k \geq 0$.

$x_k = a \cdot 3^k \Rightarrow x_{k+1} + (-2) \cdot a \cdot 3^k = 3^k$

$\Rightarrow x_{k+1} = 3^k + 2 \cdot a \cdot 3^k = 3^k (2a+1)$

$\Rightarrow a \cdot 3^{k+1} = 3^k (2a+1)$

$\Rightarrow 3a \cdot 3^k = (2a+1) 3^k \ \big| : 3^k \neq 0$

$\Rightarrow 3a = 2a + 1 \Rightarrow a = 1.$

2)

b) gen. sol. of $x_{k+1} - 2x_k = 3^k$

$r - 2 = 0 \Rightarrow r = 2 \longmapsto 2^k.$

$\Rightarrow x_h = x_k = c \cdot 2^k, \ c \in \mathbb{R}.$

from a) $\Rightarrow x_p = 3^k$

$\Rightarrow x = x_h + x_p = c \cdot 2^k + 3^k, \ c \in \mathbb{R}.$

c) find sol. of IVP: $\begin{cases} x_{k+1} - 2x_k = 3^k \\ x_0 = 0 \end{cases}, k \geq 0.$

from b) $\Rightarrow$ sol: $x_k = c \cdot 2^k + 3^k, \ c \in \mathbb{R}.$

$x_0 = 0 \Rightarrow k = 0 \Rightarrow x_0 = c \cdot 2^0 + 3^0 = c + 1 = 0 \Rightarrow c = -1.$

$\Rightarrow x = -2^k + 3^k$ unique sol. of this IVP.

Ex2. a) Sol. of the form $x_k = ak + b$ of $x_{k+1} + 5x_k = -k, k \geq 0.$

$\Rightarrow a(k+1) + b + 5(ak + b) = -k$

$\Rightarrow ak + a + b + 5ak + 5b = -k$

$\Rightarrow 6ak + a + 6b = -k$

$\Rightarrow \begin{cases} 6a = -1 & \Rightarrow a = -\frac{1}{6} \\ a + 6b = 0 & \Rightarrow -\frac{1}{6} + 6b = 0 \Rightarrow 6b = \frac{1}{6} \Rightarrow 36b = 1 \Rightarrow b = \frac{1}{36}. \end{cases}$

$\Rightarrow x_k = -\frac{1}{6}k + \frac{1}{36}.$

b) gen. sol. of $x_{k+1} + 5x_k = -k.$

$r + 5 = 0 \Rightarrow r = -5 \longmapsto (-5)^k$

$\Rightarrow x_h = c \cdot (-5)^k, \ k \geq 0.$

from a) $\Rightarrow x_p = -\frac{1}{6}k + \frac{1}{36}$

$x = x_h + x_p = c \cdot (-5)^k - \frac{1}{6}k + \frac{1}{36}, \ c \in \mathbb{R}$

c) sol. of IVP: $\begin{cases} x_{k+1} + 5x_k = -k \\ x_0 = -1 \end{cases}$

$x_0 = -1 \Rightarrow k = 0 \Rightarrow x_0 = c + \frac{1}{36} = -1 \Rightarrow c = -1 - \frac{1}{36} = -\frac{37}{36}$

$x_k = -\frac{37}{36}(-5)^k - \frac{1}{6}k + \frac{1}{36}$ unique sol. of the IVP.

(2) Ex1: Gen. sol: a) $x_{k+2} - 6x_{k+1} + 9x_k = 0$.

$$r^2 - 6r + 9 = 0$$

$$\Delta = 36 - 36 = 0 \Rightarrow r_1 = r_2 = 3 \in \mathbb{R}$$

$$\longmapsto (3)^k, \quad k \cdot (3)^k$$

$$\Rightarrow x = c_1 \cdot 3^k + c_2 \cdot k \cdot 3^k; \quad c_1, c_2 \in \mathbb{R}.$$

b) $x_{k+2} + 4x_{k+1} + 4x_k = 0$.

$$r^2 + 4r + 4 = 0$$

$$\Delta = +0 \Rightarrow r_1 = r_2 = -2$$

$$\longmapsto (-2)^k, \quad k \cdot (-2)^k$$

$$\Rightarrow x = c_1 \cdot (-2)^k + c_2 \cdot k \cdot (-2)^k$$

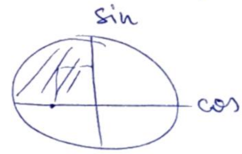
c) $x_{k+2} + x_{k+1} + x_k = 0$

$$r^2 + r + 1 = 0$$

$$\Delta = -3 < 0 \Rightarrow r_{1,2} = \frac{-1 \pm i\sqrt{3}}{2}$$

Take $r_1 = \frac{-1 + i\sqrt{3}}{2} = \rho(\cos\theta + i \cdot \sin\theta) = 1 \cdot \left(\cos\frac{2\pi}{3} + i \cdot \sin\frac{2\pi}{3}\right)$.

$$\rho = \sqrt{\left(-\tfrac{1}{2}\right)^2 + \left(\tfrac{\sqrt{3}}{2}\right)^2} = \sqrt{\tfrac{1}{4} + \tfrac{3}{4}} = \sqrt{\tfrac{4}{4}} = 1.$$



$$\left.\begin{array}{l} \cos\theta = -\frac{1}{2} < 0 \\ \sin\theta = \frac{\sqrt{3}}{2} > 0 \end{array}\right\} \Rightarrow \theta \in C_{II} \Rightarrow \theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\longmapsto \cancel{(r_1)^k, \; k(r_1)^k} \quad \mathrm{Re}(\alpha + i\beta)^k, \; \mathrm{Im}(\alpha + i\beta)^k$$

$$\parallel$$

$$1^k \cdot \left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)^k = \underbrace{\cos\frac{2k\pi}{3}}_{\mathrm{Re}} + i\underbrace{\sin\frac{2k\pi}{3}}_{\mathrm{Im}}$$

$$\Rightarrow x_k = c_1 \cdot \cos\frac{2k\pi}{3} + c_2 \cdot \sin\frac{2k\pi}{3}; \quad c_1, c_2 \in \mathbb{R}.$$

Ex2: Expression of the Fibonacci seq: $x_{k+2} = x_{k+1} + x_k$, $x_0 = 0$, $x_1 = 1$

$$r^2 = r + 1 \Rightarrow r^2 - r - 1 = 0$$

$$\Delta = 5 \Rightarrow r_{1,2} = \frac{1 \pm \sqrt{5}}{2} \in \mathbb{R}.$$

$x_k = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^k + \left(-\frac{1}{\sqrt{5}}\right)\left(\frac{1-\sqrt{5}}{2}\right)^k$

↳ unique sol. of the IVP.

$$\longmapsto (r_1)^k, \; \cancel{k} \cdot (r_2)^k$$

$$\Rightarrow x_k = c_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^k + c_2 \cdot \left(\frac{1-\sqrt{5}}{2}\right)^k; \quad c_1, c_2 \in \mathbb{R}.$$

$$x_0 = 0 \Rightarrow k = 0 \Rightarrow x_0 = c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$$

$$x_1 = 1 \Rightarrow k = 1 \Rightarrow x_1 = c_1 \cdot \frac{1+\sqrt{5}}{2} + c_2 \cdot \frac{1-\sqrt{5}}{2} = 1$$

$$\Rightarrow -c_2 \cdot \frac{1+\sqrt{5}}{2} + c_2 \cdot \frac{1-\sqrt{5}}{2} = 1 \Rightarrow -c_2 \cdot \frac{\sqrt{5}}{2} - c_2 \cdot \frac{\sqrt{5}}{2} = 1 \Rightarrow -\sqrt{5}\,c_2 = 1$$

$$\Rightarrow c_1 = \frac{1}{\sqrt{5}}$$

$$\Rightarrow c_2 = -\frac{1}{\sqrt{5}}$$

Ex3. Find a LHDE with c.c. whose gen. sol. is:

a) $x_k = \frac{c_1}{2^k} + \frac{c_2}{3^k}$; $c_1, c_2 \in \mathbb{R}$

$r_1 \longmapsto r_1^k$; $\frac{1}{2} \longmapsto \left(\frac{1}{2}\right)^k$

$r_2 \longmapsto r_2^k$; $\frac{1}{3} \longmapsto \left(\frac{1}{3}\right)^k$

$\Rightarrow$ char. eq: $\left(r-\frac{1}{2}\right)\left(r-\frac{1}{3}\right)=0$

$\Rightarrow r^2 - \frac{5}{6}r + \frac{1}{6} = 0$

$\Rightarrow x_{k+2} - \frac{5}{6}x_{k+1} + \frac{1}{6}x_k = 0.$

b) $x_k = c_1 \cdot Re(i^k) + c_2 \cdot Im(i^k)$; $c_1, c_2 \in \mathbb{R}$; x_5, x_6, x_7?

$r = \pm i \Rightarrow r^2+1=0.$

$\not\Rightarrow (r-i)(r+i)=0$

$\Rightarrow r^2 + \cancel{ri} - \cancel{ri} - i^2 = 0$

$\Rightarrow r^2+1=0$

$x_{k+2} + x_k = 0$

$x_5 = c_1 \cdot Re(i^5) + c_2 \cdot Im(i^5) = c_1 \cdot \overset{0}{Re(i)} + c_2 \cdot \overset{1}{Im(i)} =$
$k=5$
$= 0 + c_2 \cdot 1 = c_2.$

$x_6 = c_1 \cdot Re(i^6) + c_2 \cdot Im(i^6) = c_1 \cdot Re(i^2) + c_2 \cdot Im(i^2)$
$k=6$
$= +c_1 + c_2 \cdot 0 = +c_1.$ $-c_1$?

$x_7 = c_1 \cdot Re(i^7) + c_2 \cdot Im(i^7) = c_1 \cdot Re(i^3) + c_2 \cdot Im(i^3)$
$k=7$
$= c_1 \cdot 0 + c_2 \cdot (-1) = -c_2.$

$\boxed{4}$. $\ell(r) = (r-1)^2 \cdot (r+5)$

$\mathcal{L} = (D-1)^2 \cdot (D+5)$

$\mathcal{L}(x) = ?$ for $x \in \{e^{-t}, t \cdot e^t, t^2 \cdot e^t\}$

• e^{-t}: $(D+5)(e^{-t}) = (e^{-t})' + 5 \cdot e^{-t} = \ldots$

$(D-1)(e^{-t}) = (e^{-t})' - 1 \cdot e^{-t} = \ldots$

$\mathcal{L}(e^{-t}) = (D-1)^2 \cdot \underbrace{(D+5)(e^{-t})}$

$= (D-1)^2 \cdot (\ldots)$

$= 4 \ldots$

$\boxed{3}$ Reduction method (linear planar sys. w c.c.)

$$\begin{cases} x' = a_{11}x + a_{12}y \\ y' = a_{21}x + a_{22}y \end{cases} \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in \mathcal{M}_n(\mathbb{R})$$

if $a_{21} = a_{12} = 0 \Rightarrow$ uncoupled $\Rightarrow$

$\Rightarrow$ sol: $\begin{cases} x = c_1 \cdot e^{a_{11} \cdot t} \\ y = c_2 \cdot e^{a_{22} \cdot t} \end{cases}$

ex: $e^{t \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}$; $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$\begin{cases} x' = y \\ y' = -x \end{cases} \Rightarrow x'' = y' \Rightarrow x'' = -x \Rightarrow x'' + x = 0$$

$r^2 + 1 = 0 \Rightarrow r = \pm i$.

$\mapsto e^{0 \cdot t} \cdot \sin t,\ e^{0 \cdot t} \cdot \cos t$

$\Rightarrow x(t) = c_1 \cdot \cos t + c_2 \cdot \sin t$

$y(t) = x'(t) = -c_1 \cdot \sin t + c_2 \cdot \cos t$

IVP: $\begin{cases} x' = y \\ y' = -x \\ x(0) = 1 \Rightarrow c_1 = 1 \\ y(0) = 0 \Rightarrow c_2 = 0 \end{cases} \Rightarrow \varphi(t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$ or $\begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$

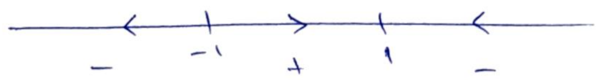
$\Rightarrow x(t) = \cos t$

$y(t) = -\sin t$

$$\Rightarrow e^{t \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$$

$\boxed{4}$ Phase portrait (d.s. associated to a scalar diff. eq.)

$f(x) = 1 - x^2 = 0 \Rightarrow x_{1,2} = \pm 1$.



x	$-\infty$		-1		1		∞
$f(x)$		$-$	0	$+$	0	$-$	

$\Rightarrow$ orbits = intervals. $\Rightarrow (-\infty, -1), \{-1\}, \{1\}, (-1, 1), (1, \infty)$

$\boxed{5}$. Attractor for continuous / discrete dyn. sys.

Continuous DS: $\boxed{x' = f(x)}$ $\rightarrow$ system, equation

x^* equilibrium point $(f(x) = 0)$ $\Rightarrow \begin{cases} f'(x^*) < 0 \Rightarrow \text{attr.} \\ f'(x^*) > 0 \Rightarrow \text{rep.} \\ f'(x^*) = 0 \Rightarrow \text{lin. fails} \end{cases}$

Discrete DS: $\boxed{x_{k+1} = f(x_k)}$ $\rightarrow$ function, map

x^* fixed point $(f(x) = x)$ $\Rightarrow$ $|f'(x^*)| < 1 \Rightarrow x^*$ attractor.

$|f'(x^*)| > 1 \Rightarrow x^*$ unstable fixed point.

$\underline{\text{Ex. 1}}$ $x_{k+1} = x_k + h(x_k^2 + 5x_k + 6) =: F(x)$ (DISCRETE DYN. SYSTEM)

a) fixed points? $F(x) = x$

$$\Rightarrow h(x^2 - 5x + 6) = 0$$

$$\Rightarrow (x-2)(x-3) = 0 \Rightarrow x^* = 2,\ x^* = 3$$

$$F'(x) = 1 + h(2x - 5)$$

$F'(2) = 1 + h\cdot(-1) = 1 - h$; attractor $\Leftrightarrow |1-h| < 1 \Leftrightarrow 0 < h < 2$.

$F'(3) = 1 + \underbrace{h}_{>0} > 1,\ (\forall)\, h > 0 \Rightarrow x^* = 3$ unstable.

$\boxed{6}$ „Associated" planar system: $\theta'' + \omega^2 \sin\theta = 0$

$$\begin{cases} x = \theta \Rightarrow \underline{x' = \theta'} = y \\ y = \theta' \Rightarrow y' = \theta'' \Rightarrow \underline{y' = -\omega^2 \sin\theta = -\omega^2 \sin x}. \end{cases}$$

$\boxed{7}$ Compute $e^{\begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}}$

$$A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} = \beta \cdot \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{B}$$

$$B^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2$$

$$B^3 = B^2 \cdot B = -I_2 \cdot B = -B$$

$$B^4 = B^3 \cdot B = -B \cdot B = -B^2 = I_2$$

$$B^{2k} = (-1)^k \cdot I_2$$

$$B^{2k+1} = (-1)^k \cdot I_2$$

$$e^{\begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}} = e^{\beta \cdot B} = \sum_{k=0}^{\infty} \frac{\beta^k}{k!} B^k =$$

$$= \sum_{\substack{k=0 \\ k \text{ even}}}^{\infty} \frac{\beta^k}{k!} B^k + \sum_{\substack{k=0 \\ k \text{ odd}}}^{\infty} \frac{\beta^k}{k!} B^k$$

$$= \sum_{m=0}^{\infty} \frac{\beta^{2m}}{(2m)!} B^{2m} + \sum_{m=0}^{\infty} \frac{\beta^{2m+1}}{(2m+1)!} B^{2m+1}$$

$$= \underbrace{\left(\sum_{m=0}^{\infty} \frac{(-1)^m I_2}{(2m)!} \cdot \beta^{2m} \right)}_{\cos\beta} I_2 + \underbrace{\left(\sum_{m=0}^{\infty} \frac{(-1)^m \cdot \beta^{2m+1}}{(2m+1)!} \right)}_{\sin\beta} I_2$$

8. $x' = 1 - x^2$; $\varphi(\cdot, \eta)$ the sol. of the IVP.

a) $\varphi(t,-1) = ?$ $\varphi(t,1) = ?$

IVP: $\begin{cases} x' = 1 - x^2 = (1-x)(1+x) = 0 \Rightarrow \eta_1^* = 1, \eta_2^* = -1 \text{ equilibrium points.} \\ x(0) = \eta = -1 \end{cases}$

!

$\varphi(t,-1) \Rightarrow \eta = -1 = \eta_1^* = \text{equil.} \Longrightarrow \varphi(t,-1) = -1$

$\varphi(t,+1)$ the sol. of the IVP: $\begin{cases} x' = 1 - x^2 \\ x(0) = 1 \end{cases}$

$\boxed{\begin{matrix} \varphi(t, \eta^*) = \eta^* \\ (\Leftrightarrow) \ \eta^* \text{ equil.} \end{matrix}}$

$\underline{\eta = 1 \text{ equil.}}$ actually $\Rightarrow \boxed{\varphi(t,1) = 1}$

b) prove $-1 < \varphi(\cdot, 0) < 1$, $(\forall) t \in \mathbb{R}$.

$\varphi(\cdot, 0)$ strictly increasing, $\lim_{t \to \infty} \varphi(t,0) = 1$

$\lim_{t \to -\infty} \varphi(t,0) = -1$

$\varphi(t,0)$ sol. of IVP: $\begin{cases} x' = 1 - x^2 \\ x(0) = 0 \end{cases}$

$\overset{\|}{\varphi(t)}$

$\begin{cases} \varphi'(t) = f(\varphi(t)) \\ \varphi(0) = 0 \quad \text{regular point (not equil.)} \end{cases}$

if $\varphi(0) < \eta^*$
$\Rightarrow \varphi(t) < \eta^* \ldots$

$-1 < \underset{0}{\varphi(0)} < 1 \xrightarrow{\text{Prop 2}} -1 < \varphi(t) < 1$, $(\forall) t \in I$.

$\Rightarrow \varphi$ bounded $\xrightarrow{\text{Th.}} I = \mathbb{R} \Rightarrow \varphi : I \to \mathbb{R}$

IVP has
a unique sol.

$\begin{rcases} \varphi \text{ not cont.} \Rightarrow \varphi \text{ strictly monotone} \\ f(\varphi(t)) > 0, (\forall) t \in \mathbb{R} \end{rcases} \Rightarrow \varphi$ strictly increasing.

9. Check $H: \mathbb{R}^2 \to \mathbb{R}$, $H(x,y) = xy$ is a first integral of: $\begin{cases} \dot{x} = -x \\ \dot{y} = y \end{cases}$ using the def. of the integral.

$\varphi(t, \eta)$ flow = ? $m = 2$.

$\boxed{\text{Def} \Rightarrow (\forall)\ \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}, \varphi(t,\eta) \text{ is the sol. of the IVP}}$

(char. eq. method)

$\begin{cases} \dot{x} = -x \Rightarrow \dot{x} + x = 0 \Rightarrow r + 1 = 0 \Rightarrow r = -1 \leftrightarrow e^{-t} \Rightarrow x = c \cdot e^{-t} \Rightarrow \underline{x(t) = \eta_1 \cdot e^{-t}}. \\ \quad x(0) = \eta_1 \Rightarrow c = \eta_1 \\ \dot{y} = y \Rightarrow \dot{y} - y = 0 \Rightarrow r - 1 = 0 \Rightarrow r = 1 \\ \quad \leftrightarrow y = c \cdot e^t \\ x(0) = \eta_1 \\ y(0) = \eta_2 \quad y(0) = \eta_2 = c \Rightarrow \underline{y(t) = \eta_2 \cdot e^t} \end{cases}$

$\Rightarrow \varphi(t, \eta_1, \eta_2) = \begin{pmatrix} \eta_1 \cdot e^{-t} \\ \eta_2 \cdot e^t \end{pmatrix}$, $(\forall) t \in \mathbb{R}$

$H: \mathbb{R}^2 \to \mathbb{R}$, $H(x,y) = xy$

H not locally ct.

$\boxed{\text{Def} \Rightarrow H(\varphi(t,\eta)) = H(\eta)}$ $(\Leftrightarrow)$ $\eta_1 \cdot \cancel{e^{-t}} \cdot \eta_2 \cdot \cancel{e^t} = \eta_1 \cdot \eta_2$ true, $= H(\eta_1, \eta_2)$, $(\forall) t \in \mathbb{R}$
$(\forall) \eta \in \mathbb{R}^2$

10. Check, using the definition, that $H:\mathbb{R}^2\to\mathbb{R}$, $H(x,y)=x^2+y^2$ f.i.

! $\begin{cases}\dot{x}=-y\\ \dot{y}=x\end{cases}$ $\Rightarrow$ FLOW. ; Let $\eta=\begin{pmatrix}\eta_1\\ \eta_2\end{pmatrix}$ and IVP $\begin{cases}\dot{x}=-y\\ \dot{y}=x\end{cases}$

Reduction method.

$\dot{x}=-y \Rightarrow y=-\dot{x}=+c_1\cdot\sin t-c_2\cdot\cos t$

$\ddot{x}=-\dot{y}=-x \Rightarrow \ddot{x}+x=0 \Rightarrow r^2+1=0 \Rightarrow r=\pm i \longleftrightarrow \cos t, \sin t$

$\Rightarrow\begin{cases}x(t)=c_1\cdot\cos t+c_2\cdot\sin t\ ;\ x'(t)=-c_1\cdot\sin t+c_2\cdot\cos t\\ y(t)=c_1\cdot\sin t-c_2\cdot\cos t\end{cases}$

$x(0)=\eta_1 \Rightarrow c_1=\eta_1$

$y(0)=\eta_2 \Rightarrow -c_2=\eta_2 \Rightarrow c_2=-\eta_2$

$\Rightarrow\begin{cases}x(t)=\eta_1\cdot\cos t-\eta_2\cdot\sin t\\ y(t)=\eta_1\cdot\sin t+\eta_2\cdot\cos t\end{cases} \Rightarrow \varphi(t,\eta_1,\eta_2)=\begin{pmatrix}\eta_1\cdot\cos t-\eta_2\sin t\\ \eta_1\sin t+\eta_2\cos t\end{pmatrix}$

$\boxed{\text{Def} \Rightarrow H(\varphi(t,\eta))\overset{?}{=}H(\eta)}$

$H(\varphi(t,\eta))=(\eta_1\cos t-\eta_2\sin t)^2+(\eta_1\sin t+\eta_2\cos t)^2=$

$=\eta_1^2\cdot\cos^2 t\cancel{-2\eta_1\eta_2\sin t\cos t}+\eta_2^2\sin^2 t+$

$+\eta_1^2\sin^2 t\cancel{+2\eta_1\eta_2\sin t\cos t}+\eta_2^2\cos^2 t$

$=\eta_1^2(\sin^2 t+\cos^2 t)+\eta_2^2(\sin^2 t+\cos^2 t)$

$=\eta_1^2+\eta_2^2$

$=H(\eta)$, $(\forall)t\in\mathbb{R}$, $(\forall)\eta\in\mathbb{R}^2$.

11. Phase portrait of $\begin{cases}\dot{x}=-y\\ \dot{y}=x\end{cases}$ $H:\mathbb{R}^2\to\mathbb{R}$, $H(x,y)=x^2+y^2$ f.i.

(M1) $\Rightarrow$ orbits = level curves of H.

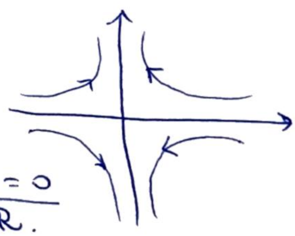
$\varphi(t,1,0)=\begin{pmatrix}\cos t\\ \sin t\end{pmatrix}$ $\Rightarrow$ trigon. sense

(M2) $f(x,y)=\begin{pmatrix}-y\\ x\end{pmatrix}$; $f(1,0)=\begin{pmatrix}0\\ 1\end{pmatrix} \Rightarrow$ up / right.

$f(0,1)=\begin{pmatrix}-1\\ 0\end{pmatrix} \Rightarrow$ left / down

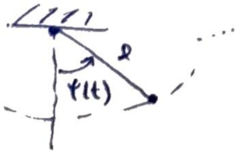
12 $\begin{cases}\dot{x}=-x\\ \dot{y}=y\end{cases}$ equil. $(0,0)$. x pos $\Rightarrow \dot{x}$ neg $\Rightarrow$ <

x neg $\Rightarrow \dot{x}$ pos $\Rightarrow$ >



13. Non-linear simple pendulum eq: $\ddot{x}+w^2\cdot\sin x=0$

Let $\varphi:\mathbb{R}\to\mathbb{R}^2$ sol. s.t. $\varphi''(t)+w^2\cdot\sin\varphi(t)=0$, $(\forall)t\in\mathbb{R}$.



$\boxed{14}$ General sol. of a separable eq: $\frac{dy}{dx} = g_1(x) \cdot g_2(y)$

$\Rightarrow$ 1) separate var: $\frac{dy}{g_2(y)} = g_1(x)\,dx$

2) integrate: $G_2(y) = \int \frac{dy}{g_2(y)} = \int g_1(x)\,dx$

$G_1(x) = \int g_1(x)\,dx$

$\Rightarrow$ Sol: $G_2(y) = G_1(x) + C$, $C \in \mathbb{R}$ $\rightarrow$ simplify etc.

! $\boxed{15}$ $\ddot{x} + w^2 \sin x = 0$

a) f.i. ..... $H(x,y) = y^2 - 2w^2 \cos x$ ...

check .... ✓

b) $\varphi^*(t) = 0$ stable equil.?

$\hookrightarrow$ sol. of $\ddot{x} + w^2 \sin x = 0$

$\hookrightarrow$ corresponds to $(0,0)$ equil. point of $\begin{cases} \dot{x} = y \\ \dot{y} = -w^2 \sin x \end{cases}$

<u>Lin method</u>: $f(x,y) = \begin{pmatrix} y \\ -w^2 \sin x \end{pmatrix}$

$Jf(x,y) = \begin{pmatrix} 0 & 1 \\ -w^2 \cos x & 0 \end{pmatrix}$

$Jf(0,0) = \begin{pmatrix} 0 & 1 \\ -w^2 & 0 \end{pmatrix}$ $\Rightarrow$ ... $\Rightarrow \lambda = \pm i w$ $\Rightarrow$ CENTER, stable?

! $(0,0)$ hyp? No, because $Re(\lambda) = 0$. $\Rightarrow$ lin. fails.

$\Rightarrow$ <u>Try H</u>: $\begin{cases} \lambda = \pm i w \text{ the eigenvalues of the linearisation around } (0,0) \\ \exists H: \mathbb{R}^2 \rightarrow \mathbb{R} \text{ first integral} \end{cases}$

$\xRightarrow{\text{Theorem}}$ $(0,0)$ stable.

$\boxed{16}$ Euler numerical formula to find $y_1, \dots, y_n$

$\rightarrow$ step1: $m_0 = g(x_0, y_0)$ slope of dir. field in (x_0, y_0)

$m_0 = \frac{c_1}{c_2} = \frac{y_1 - y_0}{x_1 - x_0}$ $\Rightarrow y_1 = y_0 + m_0(x_1 - x_0)$

$\underline{y_1 = y_0 + (x_1 - x_0) \cdot g(x_0, y_0)}$

$\rightarrow$ step2: $m_1 = g(x_1, y_1)$

$\Rightarrow m_1 = \dots \Rightarrow \underline{y_2 = y_1 + (x_2 - x_1) \cdot g(x_1, y_1)}$

⋮

$\rightarrow$ step k: $\boxed{y_{k+1} = y_k + (x_{k+1} - x_k) \cdot g(x_k, y_k)}$

! Particular: $h = ct$:

$$\boxed{\begin{cases} x_{k+1} = x_k + h & \boxed{h = \frac{x^* - x_0}{n}} \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}}$$

x_0, y_0 given

$k = \overline{0, n-1}$.

$\underline{Ex1}$ Euler num. formula w ct. stepsize $h>0$ for IVP: $\begin{cases} y' = 1 + xy^2 \\ y(0) = 0 \end{cases}$

$h = 0.1$. Nr. steps to reach $x^* = 1$? Approx. values $\varphi(0.1)$? $\varphi(0.2)$?

! $h = 0.1 = \dfrac{x^* - x_0}{m} \Rightarrow \dfrac{1}{10} = \dfrac{1 - x_0}{m}$

$y' = 1 + xy^2 = g(x,y)$

$y(x_0) = y_0 \Rightarrow x_0 = 0,\ y_0 = 0.$

a) $h>0$ given

$x_0 = y_0 = 0$

Fix $x^* \Rightarrow h = \dfrac{x^* - x_0}{m} = 0.1. \Leftrightarrow \cancel{0.01}\ \dfrac{1}{10} = \dfrac{1-0}{m} \Rightarrow m = 10.$

$g(x,y) = y' = 1 + xy^2.$

$g: \mathbb{R}^2 \to \mathbb{R}$, g is a C^1 function

b) $m = 10$

$$\begin{cases} x_{k+1} = x_k + h = x_k + 0.1 \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{cases}$$

$x_{k+1} = x_k + 0.1.$

$x_0 = 0$

$x_1 = x_0 + 0.1 = 0.1. = 1 \cdot 0.1$

$x_2 = x_1 + 0.1 = 0.1 + 0.1 = 0.2 = 2 \cdot 0.1$

$\vdots$

$x_k = \dots = k \cdot 0.1 = \dfrac{k}{10} = k \cdot h$

$y_{k+1} = y_k + 0.1 \cdot g(x_k, y_k)$

$y_0 = 0$

$y_1 = y_0 + 0.1 \cdot (1 + \overbrace{x_0 \cdot y_0^2}^{0}) = 0 + 0.1 = 0.1$

$y_2 = y_1 + 0.1 \cdot (1 + x_1 \cdot y_1^2) = 0.1 + 0.1 \cdot (1 + (0.1 \cdot (0.1)^2) = \dots = 0.2001.$

$\cancel{g(x_0, y_0) = 1}$

$\cancel{g(x_1, y_1) =}$

? $\underline{Ex2}$: IVP: $\begin{cases} y' = y \\ y(0) = 1 \end{cases}$, $\varphi(x) = e^x$ unique sol. $\Rightarrow x_0 = 0,\ y_0 = 1.$

a) Euler num. formula with ct. step size $h>0$ and on $[0, x^*]$, x^* fixed.

$$\begin{cases} h = \dfrac{x^* - x_0}{m} \Rightarrow m = \dfrac{x^* - \cancel{x_0}}{h};\ g(x,y) = y \\ x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) = y_k + h \cdot y_k = (h+1) y_k \\ y_0 = 1,\ y_1 = 1,\ \dots,\ y_k = 1,\ x_0 = 0. \end{cases}$$

b) $y_k = (1+h)^k$ (induction)

$x_k = k \cdot h$

c) x^* fixed. $m = \dfrac{x^*}{h}$; $\lim\limits_{m} y_m = e^{x^*}$?

$\boxed{y_m \approx \varphi(x_m)} = \varphi(x^*) = e^{x^*}$

$y_m = \left(1 + \underbrace{\dfrac{x^*}{m}}_{h}\right)^m$ / lim

$\Rightarrow \lim\limits_{m} y_m = \lim\limits_{m} \left(1 + \dfrac{x^*}{m}\right)^m = e^{x^*}.$